

SUNNY KUMAR

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Linkedin

Github

GFG

Portfolio

SUMMARY

- Aspiring Software Developer with strong knowledge of Java, Spring Boot, MySQL, and Object-Oriented Programming. Passionate about building scalable backend applications, solving complex problems, and continuously learning new technologies. Seeking an opportunity to contribute to organizational success while enhancing my technical and professional skills.

EDUCATION

Indian Institute of Information Technology, Kalyani

2022 – 2026

B.Tech - Computer Science and Engineering - CGPA: 6.76

Kalyani, West Bengal

PROJECTS

Hospital Management System

Java, Spring Boot, MySQL, REST API, Hibernate/JPA, Spring Security, JWT

2025

- Developed a backend system to manage hospital operations including patient records, doctor details, appointments, and billing.
- Built RESTful APIs using Spring Boot with full CRUD functionality and integrated MySQL using Hibernate/JPA for efficient data handling.
- Implemented secure authentication and role-based access control (Admin, Doctor, Patient) using JWT and Spring Security.
- Applied input validation (@Valid) and centralized exception handling to improve API reliability and maintainability.

Elliptic Curve Cryptography on FPGA

Verilog HDL, Spartan-7 FPGA, Finite Field Arithmetic, Xilinx Vivado, Python

2025

- Implemented hardware-based Elliptic Curve Cryptography (ECC) on Spartan-7 FPGA using Verilog HDL for secure encryption.
- Optimized point addition and scalar multiplication operations to improve hardware efficiency and reduce latency.
- Validated cryptographic computations against a Python reference model, achieving **90% computational accuracy**.
- Performed synthesis, simulation, and timing analysis using Xilinx Vivado to ensure design correctness and performance.

Musify – Music Streaming Web Application

Spring Boot, React.js, MongoDB, JWT, Tailwind CSS, Cloudinary

2024

- Developed a full-stack music streaming platform with Spring Boot backend and React.js frontend supporting playlists, albums, and media uploads.
- Implemented secure authentication and role-based authorization using Spring Security and JWT.
- Integrated Cloudinary for scalable media storage and optimized REST APIs for efficient data handling.

TECHNICAL SKILLS

Programming Languages: Java, JavaScript, SQL, HTML, CSS

Frontend Development: React.js, Tailwind CSS

Backend Development: Spring Boot, REST APIs, JWT Authentication

Databases: MySQL

Tools Platforms: Git, GitHub, Postman, VS Code

ACHIEVEMENTS

- Solved **350+** Data Structures and Algorithms problems across multiple coding platforms.
- Solved Data Structures and Algorithms problems on LeetCode.
- Strong understanding of Object-Oriented Programming (OOP) concepts in **Java**.